LESSON PLAN

EMI/EMC (14)

Class	Air Agniveer Apprentice (Engineering)	**Lesson Plan No.**	12
Total No. of Lesson Plans	46	**Time Allotted for the Session**	80 min

PART A: GUIDELINES / GENERAL INSTRUCTIONS FOR INSTRUCTORS

1. Topic	Mixed inertial system, velocity damped inertial system, damping by Doppler velocity
2. Total No. of Sessions for this Topic	01 / 01
3. Sequence of this Session	01 / 01
4. Syllabus Extract	Mixed inertial system, velocity damped inertial system, damping by Doppler velocity. Fix monitored inertial systems, stellar monitored inertial system.
5. Specific Objectives	14.1.1.1 Trainee will be able to explain and define a position on the earth.

6. List of Training Aids

Ser	Description	Remarks
(a)	BR/Text Book/Docket	Refer to the master syllabus reference list
(b)	Wall Charts	Nil
(c)	Films/Duration	Nil
(d)	PPT	Objective: teach the syllabus extract for this lesson
(e)	Models	Nil
(f)	CBT	Nil

7. Practical in Lab/Equipment Rooms/Simulators	Review of INS block diagrams and signal flow charts; Instructor supervision required for diagram interpretation.
8. Existing Knowledge of the Class	As per course prerequisites

PART B: DEVELOPMENT OF LESSON

9. Checking of Previous Knowledge / Introduction	Time: 10 min

(a) What is the primary function of an Inertial Navigation System?
(b) Define the term 'damping' in the context of INS velocity errors.
(c) List two methods used to monitor inertial system accuracy.

10. Development of Main Lesson

Sub	Content	Method	Trg Aids	Time
(a)	• Mixed inertial system definition • Velocity damped inertial system operation • Damping by Doppler velocity principles • Integration of Doppler radar data • Error correction mechanisms	L & Ds	PPT, IB	15 min
(b)	• Fix monitored inertial systems overview • Stellar monitored inertial system operation • Comparison of fix and stellar monitoring • Advantages of external monitoring • Limitations of monitoring techniques	L & Ds	PPT, PDF	20 min
(c)	• Summary of INS damping methods • Role of Doppler in velocity damping • Integration of fix and stellar data • Future trends in INS monitoring • Operational implications for aircrew	Practical	IB, PDF	15 min

11. Summary and Evaluation

Method	Trg Aids	Time
Q & A	IB	15 min

(a) Explain how Doppler velocity damping reduces inertial navigation errors.
(b) Differentiate between a fix monitored system and a stellar monitored system.
(c) Describe the operational benefit of using a mixed inertial system.

12. Home Assignment and Follow Up	Prepare a brief note on how the concepts of mixed inertial systems and velocity damping will be applied in the upcoming lesson on aircraft stability (LP 02).
Time	5 min

PART C: PREPARATION FOR TOMORROW (PFT) DIRECTIVES FOR TRAINEES

Ser	Syllabus for PFT	References	Related LPs
1	Brief introduction to aircraft stability. (Static and dynamic) explanation of longitudinal stability, lateral stability, directional stability	Refer to the master syllabus reference list	LP 13

PART D: PERSONAL EXPERIENCES (PE) OF INSTRUCTOR

Ser	Item	Remarks
(a)	Adequacy of time	
(b)	Suitability of training material/aids and additional aids required	
(c)	Doubts/intelligent questions raised by trainees	
(d)	Difficulties faced by instructor	
(e)	Other experience	

Instructor	**HoF (E/L)**
Signature: _____	Signature: _____

PART E: REVIEW SHEET

Date	Instructor's Comment	HOF (E/L)	Oi/C TDEC Remarks

Prepared by / Approved by

Prepared by	Approved by
Name, Rank, P No.	Name, Rank, P No.
Nov 26	Nov 26

Queries for Instructor

1. Confirm the master syllabus reference book/BR for this subject.
2. Confirm the class existing knowledge and prerequisites.
3. Confirm the session duration and sequence; 80 min and 01/01 were assumed.
4. Prepared-by/approved-by names, ranks, P numbers and dates must be filled before approval.
5. Confirm any training aids available beyond L & Ds, IB, PDF and PPT.

Timing Check

Introduction 10 + Main 50 + Summary 15 + Home 5 = 80 min